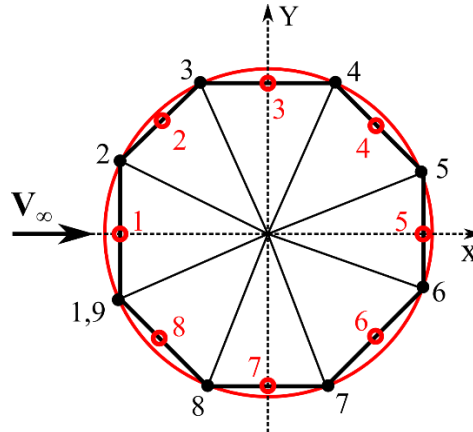


## Tutorial Problem: Source Panel Method

To calculate the pressure coefficient of a circular cylinder of unit radius, Find the value  $C_{p,2}$  using Source Panel Method.

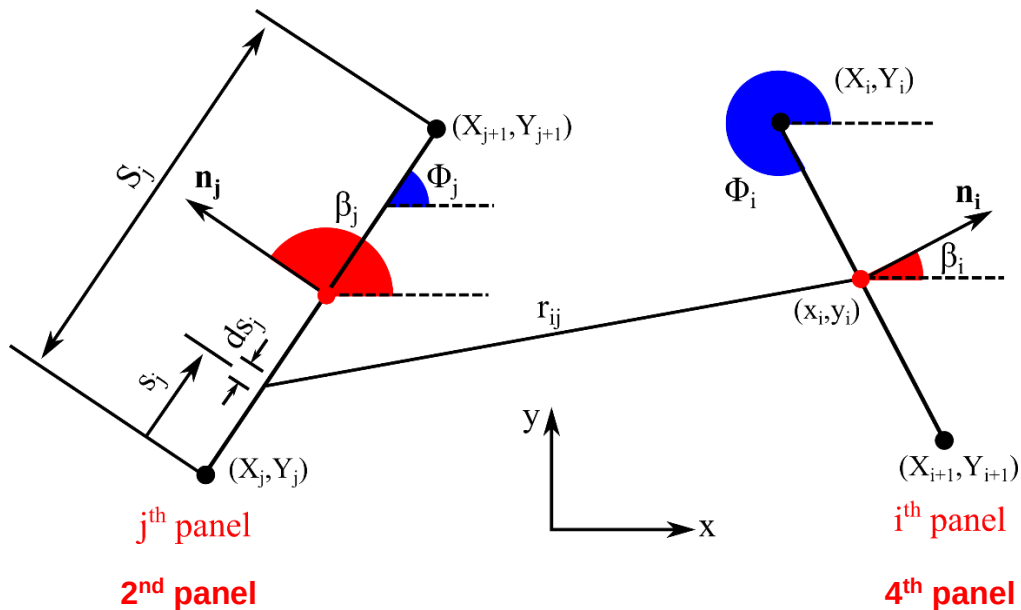


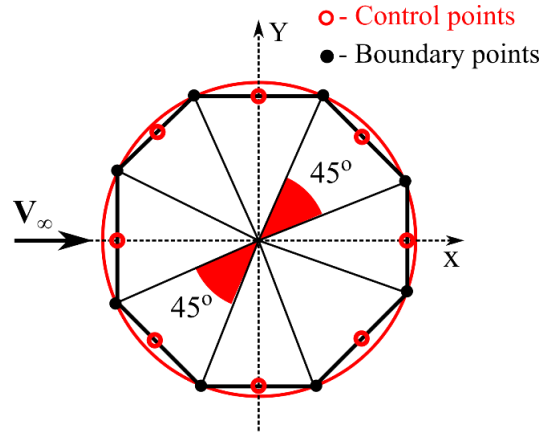
### Solution:

- $(x_i, y_i)$  are the coordinates of the control point of the  $i^{\text{th}}$  panel and  $(x_j, y_j)$  are the running coordinates over the entire  $j^{\text{th}}$  panel.
- The coordinates of the boundary points for the  $i^{\text{th}}$  panel are  $(X_i, Y_i)$  and  $(X_{i+1}, Y_{i+1})$ .
- The coordinates of the boundary points for the  $j^{\text{th}}$  panel are  $(X_j, Y_j)$  and  $(X_{j+1}, Y_{j+1})$ .
- $\beta$  is the angle between the freestream vector and the outward normal vector.
- $\Phi$  is the angle from the positive x-axis to the inside surface of the panel.
- $\delta$  is the angle from the positive x-axis to an outward normal vector of the panel.

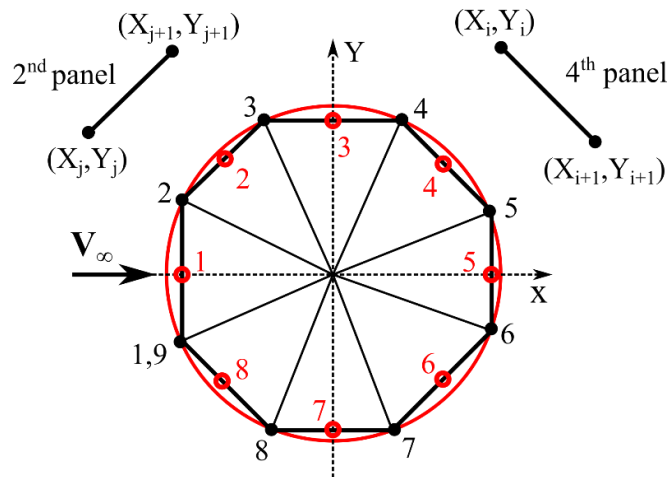
$$\beta = \phi + (\pi/2) - \alpha$$

$$\delta = \phi + (\pi/2)$$





- Dividing the circle into 8 segments, we get the angle between the boundary points  $360^\circ/8 = 45^\circ$  (angle between two boundary points).
- The coordinates of the 2<sup>nd</sup> and 4<sup>th</sup> panel can be calculated as follows:



**4<sup>th</sup> panel:**

$$X_{i+1} = r \cos \theta = 1. \cos \left( 22.5^\circ \times (\pi/180) \right) = 0.9239$$

$$Y_{i+1} = r \sin \theta = 1. \sin \left( 22.5^\circ \times (\pi/180) \right) = 0.3827$$

$$X_i = r \cos \theta = 1. \cos \left( 67.5^\circ \times (\pi/180) \right) = 0.3827$$

$$Y_i = r \sin \theta = 1. \sin \left( 67.5^\circ \times (\pi/180) \right) = 0.9239$$

**2<sup>nd</sup> Panel:**

$$X_j = r \cos \theta = 1. \cos \left( 157.5^\circ \times (\pi/180) \right) = -0.9239$$

$$Y_j = r \sin \theta = 1. \sin \left( 157.5^\circ \times (\pi/180) \right) = 0.3827$$

$$X_{j+1} = r \cos \theta = 1. \cos \left( 112.5^\circ \times (\pi/180) \right) = -0.3827$$

$$Y_{j+1} = r \sin \theta = 1. \sin \left( 112.5^\circ \times (\pi/180) \right) = 0.9239$$

The integral  $I_{ij}$  is evaluated at the 4<sup>th</sup> panel control point and the integral is taken over the complete 2<sup>nd</sup> panel.

$$I_{ij} = \int_j \frac{\partial}{\partial n_i} (\ln r_{ij}) ds_j$$

$$r_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$

$$\frac{\partial}{\partial n_i} (\ln r_{ij}) = \frac{1}{r_{ij}} \frac{\partial r_{ij}}{\partial n_i}$$

$$I_{ij} = \int_j \frac{\partial}{\partial n_i} (\ln r_{ij}) ds_j = \int_j \frac{1}{r_{ij}} \frac{\partial r_{ij}}{\partial n_i} ds_j = \int_j \frac{\left[ (x_i - x_j) \frac{\partial x_i}{\partial n_i} + (y_i - y_j) \frac{\partial y_i}{\partial n_i} \right]}{\left[ (x_i - x_j)^2 + (y_i - y_j)^2 \right]} ds_j$$

$$I_{ij} = \int_0^{\bar{S}_j} \frac{CS_j + D}{S_j^2 + 2AS_j + B} ds_j$$

**(Detailed derivation can be found in the Appendix)**

Where,

$$A = -(x_i - X_j) \cos \Phi_j - (y_i - Y_j) \sin \Phi_j$$

$$B = (x_i - X_j)^2 + (y_i - Y_j)^2$$

$$C = \sin(\Phi_i - \Phi_j)$$

$$D = (y_i - Y_j) \cos \Phi_i - (x_i - X_j) \sin \Phi_i$$

$$\bar{S}_j = \sqrt{(X_{j+1} - X_j)^2 + (Y_{j+1} - Y_j)^2}$$

$$E = \sqrt{B - A^2} = (x_i - X_j) \sin \Phi_j - (y_i - Y_j) \cos \Phi_j$$

**From any standard table of integrals:**

$$I_{i,j} = \frac{C}{2} \ln \left( \frac{\bar{S}_j^2 + 2A\bar{S}_j + B}{B} \right) + \frac{D - AC}{E} \left( \tan^{-1} \left( \frac{\bar{S}_j + A}{E} \right) - \tan^{-1} \left( \frac{A}{E} \right) \right)$$

**Assuming a unit radius for the cylinder, we see that**

$$X_j = -0.9239; X_{j+1} = -0.3827; Y_j = 0.3827$$

$$Y_{j+1} = 0.9239; \Phi_i = 315^\circ; \Phi_j = 45^\circ; x_i = 0.6533$$

$$y_i = 0.6533$$

**Hence, substituting these numbers into the above formulas, we obtain**

$$A = -1.3065; B = 2.5607; C = -1; D = 1.3065$$

$$\bar{S}_j = 0.7654; E = 0.9239$$

Inserting the above values, we obtain

$$I_{4,2} = 0.4018$$

## Appendix

From logarithmic and exponential functions,

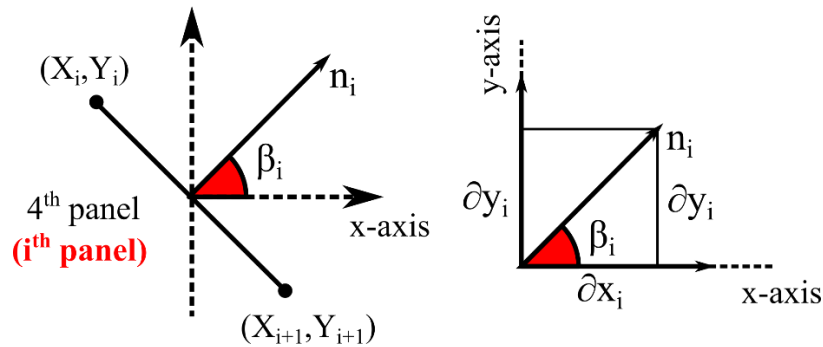
- $\frac{d}{dx}(e^x) = e^x$
- $\frac{d}{dx} \ln|x| = \frac{1}{x}$
- $\frac{d \ln(x)}{dx} = \frac{1}{x} \frac{dx}{dx} = \frac{1}{x}$
- $\frac{d[\ln(F(x))]}{dx} = \frac{1}{F(x)} \frac{dF(x)}{dx}$

Using the above function, we get

$$\begin{aligned} \frac{\partial}{\partial n_i} (\ln r_{ij}) &= \frac{1}{r_{ij}} \frac{\partial r_{ij}}{\partial n_i} = \frac{1}{r_{ij}} \frac{\partial}{\partial n_i} \left[ (x_i - x_j)^2 + (y_i - y_j)^2 \right]^{1/2} \\ \frac{\partial r_{ij}}{\partial n_i} &= \frac{1}{2} \left[ (x_i - x_j)^2 + (y_i - y_j)^2 \right]^{-1/2} \left[ 2(x_i - x_j) \frac{\partial x_i}{\partial n_i} + 2(y_i - y_j) \frac{\partial y_i}{\partial n_i} \right] \\ \frac{1}{r_{ij}} \frac{\partial r_{ij}}{\partial n_i} &= \frac{2 \left[ (x_i - x_j) \frac{\partial x_i}{\partial n_i} + (y_i - y_j) \frac{\partial y_i}{\partial n_i} \right]}{2 \left[ (x_i - x_j)^2 + (y_i - y_j)^2 \right]^{1/2} \left[ (x_i - x_j)^2 + (y_i - y_j)^2 \right]^{1/2}} \end{aligned}$$

Note:  $a^b a^c = a^{b+c}$

$$\frac{1}{r_{ij}} \frac{\partial r_{ij}}{\partial n_i} = \frac{\left[ (x_i - x_j) \frac{\partial x_i}{\partial n_i} + (y_i - y_j) \frac{\partial y_i}{\partial n_i} \right]}{\left[ (x_i - x_j)^2 + (y_i - y_j)^2 \right]}$$



From the figure, we get

$$\frac{\partial x_i}{\partial n_i} = \cos \beta_i \text{ and } \frac{\partial y_i}{\partial n_i} = \sin \beta_i$$

Hence,

$$I_{ij} = \int_j \frac{[(x_i - x_j) \cos \beta_i + (y_i - y_j) \sin \beta_i]}{[(x_i - x_j)^2 + (y_i - y_j)^2]} ds_j$$

$(x_j, y_j)$  are the running coordinates over the entire  $j^{\text{th}}$  panel.

$$x_j = X_j + s_j \cos \Phi_j$$

$$y_j = Y_j + s_j \sin \Phi_j$$

$\Phi_i$  and  $\Phi_j$  are angles measured in the counterclockwise direction from the x-axis to the bottom of each panel.

$\beta_i$  in terms of  $\Phi_i$  can be written as,

$$\beta_i = \Phi_i - 90^\circ$$

$$\cos(\beta_i) = \cos(\Phi_i + 90^\circ) = -\sin(\Phi_i)$$

$$\sin(\beta_i) = \sin(\Phi_i + 90^\circ) = \cos(\Phi_i)$$

$I_{ij}$  can be written as,

$$I_{ij} = \int_j \frac{[(x_i - x_j)(-\sin \Phi_i) + (y_i - y_j)(\cos \Phi_i)]}{[(x_i - x_j)^2 + (y_i - y_j)^2]} ds_j$$

Substitute the following terms into  $I_{ij}$ , we obtain

$$x_j = X_j + s_j \cos \Phi_j$$

$$y_j = Y_j + s_j \sin \Phi_j$$

$$I_{ij} = \int_j \frac{[(x_i - (X_j + s_j \cos \Phi_j))(-\sin \Phi_i) + (y_i - (Y_j + s_j \sin \Phi_j))(\cos \Phi_i)]}{[(x_i - (X_j + s_j \cos \Phi_j))^2 + (y_i - (Y_j + s_j \sin \Phi_j))^2]} ds_j$$

**Numerator:**

$$(x_i - (X_j + s_j \cos \Phi_j))(-\sin \Phi_i) + (y_i - (Y_j + s_j \sin \Phi_j))(\cos \Phi_i)$$

**Expanding all the terms, we get**

$$-x_i \sin(\Phi_i) + X_j \sin(\Phi_i) + s_j \sin(\Phi_i) \cos(\Phi_j) + y_i \cos(\Phi_i) - y_j \cos(\Phi_i) - s_j \cos(\Phi_i) \sin(\Phi_j)$$

**Grouping the terms, we get**

$$(X_j - x_i) \sin(\Phi_i) + (y_i - Y_j) \cos(\Phi_i) + s_j [\sin(\Phi_i) \cos(\Phi_j) - \cos(\Phi_i) \sin(\Phi_j)]$$

**We know that,**

$$\sin(A - B) = \sin(A) \cos(B) - \cos(A) \sin(B)$$

**Using the above trigonometric identity,**

$$-(x_i - X_j) \sin(\Phi_i) + (y_i - Y_j) \cos(\Phi_i) + s_j \sin(\Phi_i - \Phi_j)$$

**Numerator takes the following form:**

$$Cs_j + D$$

where

$$C = \sin(\Phi_i - \Phi_j)$$

$$D = -(x_i - X_j) \sin(\Phi_i) + (y_i - Y_j) \cos(\Phi_i)$$

**Denominator:**

$$(x_i - (X_j + s_j \cos \Phi_j))^2 + (y_i - (Y_j + s_j \sin \Phi_j))^2$$

**Expanding the terms, we get**

$$(x_i - x_j)^2 + (y_i - y_j)^2 = x_i^2 - 2x_i x_j + x_j^2 + y_i^2 - 2y_i y_j + y_j^2$$

where

$$x_j = X_j + s_j \cos(\Phi_j)$$

$$y_j = Y_j + s_j \sin(\Phi_j)$$

**Plug in for  $x_j$  and  $y_j$ , we get**

$$x_i^2 - 2x_i [X_j + s_j \cos(\Phi_j)] + [X_j + s_j \cos(\Phi_j)]^2 + y_i^2 - 2y_i [Y_j + s_j \sin(\Phi_j)] + [Y_j + s_j \sin(\Phi_j)]^2$$

**By expanding,**

$$x_i^2 - 2x_i X_j - 2x_i s_j \cos(\Phi_j) + X_j^2 + 2X_j s_j \cos(\Phi_j) + s_j^2 \cos^2(\Phi_j) + y_i^2 - 2y_i Y_j - 2y_i s_j \sin(\Phi_j) + Y_j^2 + 2Y_j s_j \sin(\Phi_j) + s_j^2 \sin^2(\Phi_j)$$

**In quadratic form,**

$$s_j^2 [\sin^2(\Phi_j) + \cos^2(\Phi_j)] + 2s_j [-x_i \cos(\Phi_j) + X_j \cos(\Phi_j) - y_i \sin(\Phi_j) + Y_j \sin(\Phi_j)] + [(x_i^2 - 2x_i X_j + X_j^2) + (y_i^2 - 2y_i Y_j + Y_j^2)]$$

**Takes the form,**

$$s_j^2 + 2As_j + B$$

where,

$$A = -(x_i - X_j)\cos(\Phi_j) - (y_i - Y_j)\sin(\Phi_j)$$

$$B = (x_i - X_j)^2 + (y_i - Y_j)^2$$

**Integral, we are trying to solve**

$$I_{i,j} = \int_0^{\bar{s}_j} \frac{Cs_j + D}{s_j^2 + 2As_j + B} ds_j$$

**Using x to make integral more recognizable,**

$$I_{i,j} = \int \frac{Cx + D}{x^2 + 2Ax + B} dx$$

$$\begin{aligned} x^2 + 2Ax + B &= x^2 + 2Ax + \left(\frac{2A}{2}\right)^2 + B - \left(\frac{2A}{2}\right)^2 \\ &= (x^2 + 2Ax + A^2) + (B - A^2) \\ &= (x + A)^2 + E^2 \end{aligned}$$

**The general procedure for integrating a partial fraction of the form,**

$$\frac{Ax + B}{ax^2 + bx + c} \text{ where } b^2 - 4ac < 0$$

**We complete the square in the denominator and then make a substitution that brings the integral into the form.**

$$\int \frac{Cu + D}{u^2 + a^2} du = C \int \frac{u}{u^2 + a^2} du + D \int \frac{1}{u^2 + a^2} du$$

**Then the first integral is a logarithmic and the second is expressed in terms of  $\tan^{-1}$**

$$I_{i,j} = \int \frac{Cx + D}{x^2 + 2Ax + B} dx$$

**Use u-substitution to solve the integral**

Define  $u = x + A$

So,  $x = u - A$  and  $dx = du$

$$I_{i,j} = \int \frac{Cx + D}{x^2 + 2Ax + B} dx = \int \frac{Cx + D}{(x + A)^2 + E^2} dx = \int \frac{C(u - A) + D}{u^2 + E^2} du = \int \frac{Cu + D - AC}{u^2 + E^2} du$$

$$\int \frac{Cu + D - AC}{u^2 + E^2} du = C \int \frac{u}{u^2 + E^2} du + (D - AC) \int \frac{du}{u^2 + E^2}$$

**Second term:**

$$\int \frac{1}{u^2 + E^2} du = \frac{1}{E} \tan^{-1} \left[ \frac{u}{E} \right]$$

we know that

$$\int \frac{du}{u^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{u}{a} + \text{constant}$$

**Plug in for  $u = x + A$  and evaluate the integral,**

$$\int_0^{\bar{S}_j} \frac{1}{u^2 + E^2} du = \left( \frac{1}{E} \tan^{-1} \left( \frac{x + A}{E} \right) \right) \Bigg|_0^{\bar{S}_j} = \frac{1}{E} \left[ \tan^{-1} \left( \frac{\bar{S}_j + A}{E} \right) - \tan^{-1} \left( \frac{A}{E} \right) \right]$$

**First term:**

$$\int \frac{u}{u^2 + E^2} du$$

**Write the numerator in a form such that it is equal to the derivative of denominator,**

$$\frac{d}{du}(u^2 + E^2) = 2u \Rightarrow \frac{1}{2} \int \frac{2u}{u^2 + E^2} du$$

**Define a new variable, Z**

$$z = u^2 + E^2 \Rightarrow \frac{dz}{du} = 2u \Rightarrow dz = 2u du$$

**Plug into integral,**

$$\frac{1}{2} \int \frac{dz}{z} = \frac{1}{2} \ln(z) = \frac{1}{2} \ln(u^2 + E^2)$$

we know that,

$$\int \frac{du}{u} = \ln|u| + C$$

**Evaluate the integral from 0 to  $\bar{S}_j$**

$$\frac{1}{2} \ln(u^2 + E^2) = \left( \frac{1}{2} \ln(x^2 + 2Ax + B) \right) \Bigg|_0^{\bar{S}_j} = \frac{1}{2} \left[ \ln(\bar{S}_j^2 + 2A\bar{S}_j + B) - \ln(B) \right]$$

**We know that**

$$\ln(A) - \ln(B) = \ln(A/B)$$



$$\int_0^{\bar{S}_j} \frac{u}{u^2 + E^2} du = \frac{1}{2} \left[ \ln \left( \frac{\bar{S}_j^2 + 2A\bar{S}_j + B}{B} \right) \right]$$

**Combining the integral expressions:**

$$I_{ij} = C \int \frac{u}{u^2 + E^2} du + (D - AC) \int \frac{1}{u^2 + E^2} du$$

$$I_{ij} = \frac{C}{2} \left[ \ln \left( \frac{\bar{S}_j^2 + 2A\bar{S}_j + B}{B} \right) \right] + \frac{D - AC}{E} \left[ \tan^{-1} \left( \frac{\bar{S}_j + A}{E} \right) - \tan^{-1} \left( \frac{A}{E} \right) \right]$$

where.

$$A = -(x_i - X_j) \cos(\Phi_j) - (y_i - Y_j) \sin(\Phi_j)$$

$$B = (x_i - X_j)^2 + (y_i - Y_j)^2$$

$$C = \sin(\Phi_i - \Phi_j)$$

$$D = -(x_i - X_j) \sin(\Phi_i) + (y_i - Y_j) \cos(\Phi_i)$$

$$E = \sqrt{B - A^2}$$

**Reference:**

Anderson, J. (2011): *Fundamentals of Aerodynamics (SI units)*. McGraw hill.